

Faster R-CNN in Healthcare and Disease Detection: A Comprehensive Review

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Abstract—This review examines the applications, challenges, and prospects of Faster Region-based Convolutional Neural Networks (Faster R-CNN) in healthcare and disease detection. Through a meta-analysis of Web of Science literature from 2017 to 2024, we provide insights into the evolving landscape of Faster R-CNN in medical contexts. The algorithm can be applied to medical image analysis across various modalities, including radiography, computed tomography, magnetic resonance imaging, ultrasound, microscopy, and endoscopy. Its applications extend to optical RGB images for dermatological and surgical uses, as well as broader healthcare areas such as posture detection, medication recognition, and assistive technologies. Despite its success, Faster R-CNN faces challenges in handling subtle abnormalities, addressing class imbalance in medical datasets, ensuring result interpretability, and managing patient privacy. Our analysis reveals dominant application areas, technological advancements, and integration trends with other artificial intelligence technologies. The review highlights Faster R-CNN's significant impact on improving diagnostic accuracy and healthcare delivery while acknowledging the need for continued research to address limitations. Emerging directions include real-time disease detection and advancements in personalized medicine, underscoring Faster R-CNN's potential to further transform healthcare practices and patient outcomes.

Keywords—Faster R-CNN, healthcare, disease detection, meta-analysis, medical image

I. INTRODUCTION

In recent years, artificial intelligence (AI) has emerged as a transformative force in healthcare, enhancing various aspects of medical practice, from diagnosis and treatment planning to patient care and health management [1-3]. Faster R-CNN, as a powerful tool for object detection, has been widely used in the field of medical images [4]. Faster R-CNN, an evolution of the R-CNN family of algorithms, was initially introduced by Ren et al. in 2015 [5]. Its ability to quickly and accurately identify and localize objects within images has made it particularly well-suited for medical applications, where precision and speed are often crucial.

In the healthcare domain, Faster R-CNN has been applied to a wide range of tasks, from detecting tumors in radiological images to identifying cellular structures in microscopy [6]. By automating and enhancing the detection of abnormalities in medical images, this algorithm has the potential to improve diagnostic accuracy, reduce the workload on healthcare professionals, and ultimately lead to better patient outcomes. Moreover, its applications have expanded beyond traditional medical imaging to include areas such as posture analysis, medication recognition, and

assistive technologies for the visually impaired, demonstrating its versatility in addressing diverse healthcare challenges [7,8]. However, the integration of AI technologies like Faster R-CNN into healthcare is not without challenges. Issues such as handling small or subtle abnormalities, addressing class imbalance in medical datasets, ensuring the interpretability of results, and managing patient privacy and data security all present significant hurdles that must be overcome for widespread adoption [9].

This review provides an examination of Faster R-CNN's role in healthcare and disease detection through two main areas of work. First, we conduct a detailed bibliometric analysis of the literature from 2017 to 2024. Our analysis shows a surge in publications since 2019, with a 40-fold increase from 2017 to 2022, indicating rapidly growing interest. Keyword clustering identifies four main research themes: medical image analysis, object detection techniques, disease diagnosis, and deep learning architectures, with strong interconnections between Faster R-CNN and medical imaging applications. Analysis of evolving terminology reveals a shift from basic implementation to advanced concepts and clinical integration. The temporal mapping of research topics illustrates a clear progression from foundational work (2017-2019) to specific medical applications (2020-2022), and recently to specialized tasks and system integration (2023-2024).

The second major component of our work involves a comprehensive review of Faster R-CNN in healthcare. We begin by explaining the fundamental principles and architecture of Faster R-CNN. We then examine in detail the applications of Faster R-CNN across various medical domains, including radiography, computed tomography, magnetic resonance imaging, ultrasound, microscopy, endoscopy, and the use of optical RGB images for dermatological and surgical applications. Our exploration extends to broader healthcare applications, such as posture detection and analysis, medication recognition, food recognition for dietary monitoring, exercise form analysis, medical equipment and supply management, and assistive technology for the visually impaired.

Furthermore, we analyze current challenges in using Faster R-CNN in healthcare, including handling subtle abnormalities, addressing class imbalance in medical datasets, ensuring result interpretability, and managing patient privacy. Our review explores potential solutions and future directions, considering both technological advancements and healthcare-specific requirements. Key emerging trends include the development of multi-modal integrated systems, improvements in model generalization across diverse datasets, and advancements in explainable AI for clinical

trust. By synthesizing current knowledge, identifying trends, and examining both the potential and limitations of Faster R-CNN, this review provides a comprehensive understanding of this tool for researchers, healthcare professionals, and policymakers, highlighting its role in advancing precision medicine and personalized healthcare.

II. BIBLIOMETRIC ANALYSIS

This review presents a systematic exploration of the research landscape and developmental trajectories in Faster R-CNN technology within the healthcare domain, employing bibliometric analysis and data visualization methodologies. The Web of Science Core Collection database was queried using the search terms "Faster R-CNN" and "healthcare" or "diagnosis," with the temporal scope set from 2017 to 2024. After deleting duplicate data, a total of 367 articles were retrieved, including 279 journal articles and 89 Proceeding Papers.

Employing CiteSpace version 6.2.R4, we conducted a multi-faceted analysis of the literature, examining four key aspects to elucidate the developmental trends and research hotspots in this technological field: Chronological trends in literature production; Keyword co-occurrence network analysis; Mutant terms analysis; and time-series cluster mapping evaluation.

A. Literature Trends

The volume of published papers serves as a valuable metric for assessing the evolutionary trajectory of a research domain. Fig. 1 provides a visual representation of the annual publication output and its fluctuations in the field of Faster R-CNN applications in healthcare spanning from 2017 to 2024.

Specifically, during the period from 2017 to 2018, the publication volume remained relatively low, with only 2 papers in 2017 and 18 papers in 2018. This likely represents the early adoption phase of Faster R-CNN in healthcare applications. From 2019 to 2021, there was a significant increase in research activity. The publication volume rose sharply to 40 papers in 2019, then to 51 in 2020, and reached 67 papers in 2021. This rapid growth indicates a surge in interest and recognition of Faster R-CNN's potential in medical applications.

The years 2022 and 2023 saw further acceleration in research output, with 82 and 65 papers published respectively, reflecting the current research boom in this field. Despite only partial data for 2024 (up to October 1, 2024), there were already 42 papers published, suggesting that the high level of research activity is likely to continue.

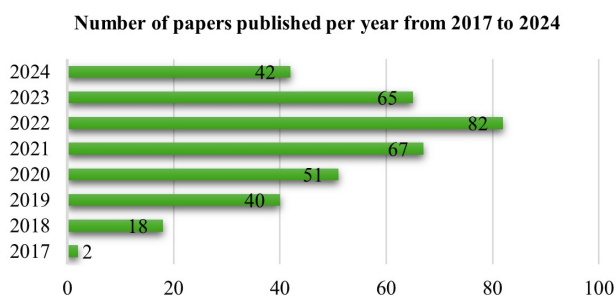


Fig. 1. Annual Publication Volume in Faster R-CNN Applications in Healthcare (2017-2024)

This trend clearly demonstrates the growing importance and recognition of Faster R-CNN in healthcare applications. The explosive growth since 2019 indicates widespread academic attention to this field, likely driven by advancements in computational power, the increasing availability of medical imaging datasets, and the pressing need for efficient and accurate diagnostic tools in healthcare.

B. Keyword co-occurrence analysis

CiteSpace's keyword co-occurrence analysis effectively illuminates research hotspots in Faster R-CNN applications in healthcare. Visualization reveals a complex network of interconnected research themes, with several dominant clusters emerging. Fig. 2 visualizes keywords based on co-occurrence frequency.

At the core of this research landscape lies "deep learning," which is the central concept connecting various research directions. Closely related and equally prominent is "artificial intelligence," indicating the fundamental role of AI techniques in this field.

"Faster R-CNN" itself stands out as a key node, directly linked to practical applications such as "object detection" and "segmentation." The presence of "convolutional neural network" underscores the technical foundation of many studies in this area. Medical applications are evident through keywords like "diagnosis" and "cancer," suggesting a significant focus on leveraging Faster R-CNN for disease detection and classification. The prominence of "classification" further reinforces this trend.

Methodological approaches are represented by terms such as "feature extraction," "transfer learning," and "machine learning," indicating a diverse toolkit employed by researchers to enhance Faster R-CNN's performance in healthcare contexts. The appearance of "computer aided diagnosis" points to the practical integration of these technologies into clinical workflows.

The temporal dimension provided by the color gradient suggests an evolution of research focus over time, with earlier years (represented by darker shades) associated with more fundamental concepts, while recent years (lighter shades) show a shift towards more specialized applications and advanced techniques. This rich tapestry of keywords reflects the multifaceted nature of Faster R-CNN research in healthcare, spanning from foundational AI concepts to specific medical applications, and highlights the field's rapid evolution towards more sophisticated and clinically relevant implementations.

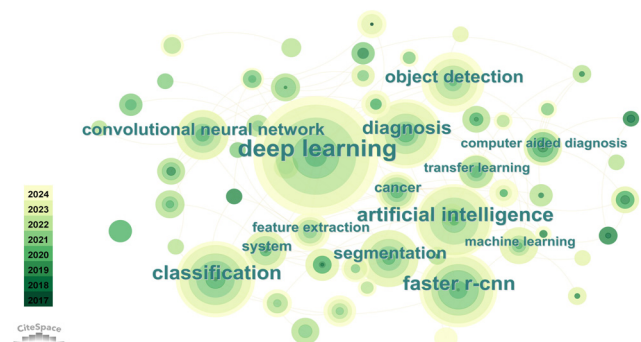


Fig. 2. Keyword Co-occurrence Network in Faster R-CNN Healthcare Research

TABLE I. TOP 10 KEYWORDS WITH THE CITATION BURSTS

Keywords	Year	Strength	Begin	End	2017 - 2024
nodules	2018	1.68	2018	2019	
computer aided diagnosis	2018	1.63	2018	2021	
features	2019	1.36	2019	2020	
automatic detection	2020	1.64	2020	2021	
neural networks	2020	1.49	2020	2022	
risk	2020	1.35	2020	2021	
disease	2021	2	2021	2022	
computed tomography	2021	1.67	2021	2022	
validation	2019	1.64	2021	2022	
system	2019	1.76	2022	2024	

C. Mutant terms

The Mutant terms analysis, generated by CiteSpace, provides insights into keyword evolution in academic literature, revealing research dynamics and trends in Faster R-CNN applications in healthcare. Table 1 presents the top 10 mutant keywords exhibiting significant citation bursts in this field.

Initially, around 2018, emphasis was placed on specific medical applications, as indicated by the emergence of terms like "nodules" and "computer aided diagnosis". This suggests early focus on applying Faster R-CNN to specific diagnostic tasks, particularly in imaging-heavy specialties like radiology.

In 2019, attention shifted towards refining the features used in Faster R-CNN models, likely indicating efforts to improve the algorithm's performance on medical imaging tasks. The years 2020-2021 saw a significant shift towards more advanced AI techniques, with "neural networks" emerging as a key term. This period also saw increased focus on "automatic detection" and "risk" assessment, suggesting a move towards more comprehensive and automated diagnostic systems.

In 2021-2022, there was a clear emphasis on specific imaging modalities, particularly "computed tomography," along with a focus on "disease" detection more broadly. The emergence of "validation" as a key term during this period indicates an increased emphasis on rigorously evaluating the performance of Faster R-CNN in clinical settings.

The most recent trend, extending into 2024, shows a focus on "system" development, suggesting a move towards more integrated and comprehensive Faster R-CNN-based diagnostic systems. These keyword mutations reflect the continuous evolution of research content and directions in this field, from early applications in specific diagnostic tasks to more sophisticated, integrated systems leveraging advanced AI techniques.

D. Clustering Timeline Mapping

The temporal clustering of keywords, visualized through CiteSpace, provides a dynamic perspective on the evolution of research themes within Faster R-CNN applications in healthcare. Fig. 3 presents this timeline graph, offering insights into the chronological development of key concepts and research foci from 2017 to 2024. In this visualization, node size corresponds to the frequency of keyword occurrence, while color gradation from dark purple to yellow represents the progression of time from 2017 to 2024.

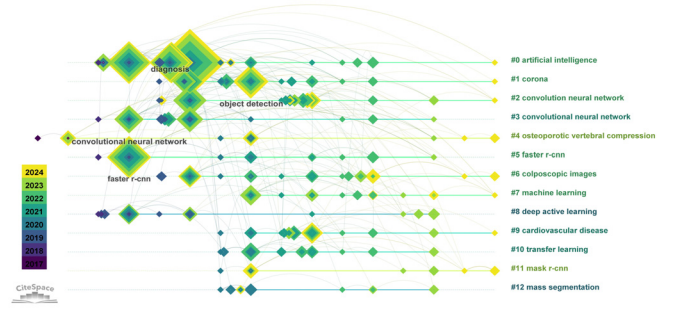


Fig. 3. Clustering timeline mapping

Interconnecting lines depict relationships between keywords, and labels on the right identify major research clusters. Early stage (2017-2019): Research primarily focused on foundational technologies, with "convolutional neural network" and "Faster R-CNN" emerging as dominant keywords. This suggests that during this period, researchers were exploring fundamental approaches to applying Faster R-CNN in medical contexts. The prominence of these terms indicates a focus on establishing the technical groundwork necessary for subsequent applications in healthcare. Mid-term developments (2020-2022): This period witnessed a diversification of research themes. "Diagnosis" and "object detection" became significant keywords, indicating the transition of Faster R-CNN into specific medical diagnostic tasks. This shift suggests a movement from theoretical foundations to practical applications in medical imaging analysis. Recent trends (2023-2024): The latest research demonstrates a trend towards specialization and application-oriented studies. The appearance of specific medical applications such as "osteoporotic vertebral compression" and "cardiovascular disease" indicates a shift towards addressing concrete clinical challenges. This trend underscores the growing integration of Faster R-CNN into specialized medical domains. Notably, "corona" related research emerges post-2020, reflecting the potential application of Faster R-CNN in addressing urgent public health crises. The introduction of "transfer learning" and "Mask R-CNN" techniques suggests ongoing efforts to improve and extend Faster R-CNN's capabilities, potentially enhancing its adaptability across various medical imaging tasks.

This timeline mapping illustrates the rapid evolution of Faster R-CNN research in healthcare, transitioning from exploratory studies of basic technologies to diverse clinical applications.

III. APPLICATIONS OF FASTER R-CNN IN HEALTHCARE

A. Architecture of Faster R-CNN

Faster R-CNN integrates two key components: the Region Proposal Network (RPN) and the Fast R-CNN detector [10]. The RPN efficiently generates regional proposals from input images using a shared convolutional backbone. These proposals are then refined and fed into the Fast R-CNN network, which performs object classification and bounding box regression. This unified approach, leveraging shared convolutional features, enables Faster R-CNN to achieve both high accuracy and real-time performance in object detection tasks. Fig. 4 shows the basic architecture of Faster R-CNN.

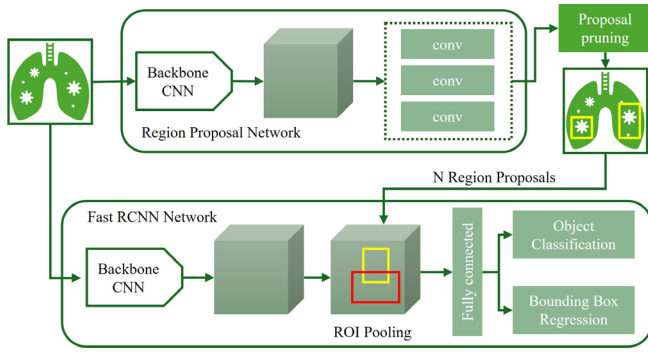


Fig. 4. Architecture of Faster R-CNN

B. Radiography (X-ray imaging)

Faster R-CNN has been widely applied to radiographic image analysis, leveraging the high-contrast, projectional nature of X-ray imaging to detect and localize various thoracic, dental, and skeletal abnormalities. Kang et al. [11] implemented a multi-object detection computer-aided diagnosis (CAD) system in the field of chest X-ray diagnostics using Faster R-CNN and RetinaNet methods. The study preprocessed chest X-ray images by matching them one-to-one with corresponding CT scan descriptions, enhancing the accuracy of data labeling. The research team trained the models to identify twelve different diagnostic labels, including pleural effusion, atelectasis, pulmonary nodules, and cardiomegaly. The application of this algorithm extends to dental inspection and detection of bone fractures. Zhu et al. [12] applied Faster R-CNN to detect and localize dental caries in periapical radiographs. Their model was trained on 200 X-ray images containing caries, with data augmentation techniques applied to expand the dataset. Abbas et al. [13] utilized Faster R-CNN for the detection and classification of lower leg bone fractures in X-ray images. Their model was trained on 50 X-ray images and achieved an overall accuracy of 94% for both fracture detection and classification.

By rapidly and accurately identifying these anomalies, Faster R-CNN has substantially enhanced the efficiency of radiologists' workflow.

C. Computed Tomography (CT)

In the domain of CT image analysis, Faster R-CNN exploits the high spatial resolution and cross-sectional capabilities of this modality to address a wide range of diagnostic tasks. Su et al. [14] applied Faster R-CNN to detect lung nodules in chest CT images. Their model was trained on 7,000 CT slices containing lung nodules. The optimized Faster R-CNN model achieved an average precision (AP) of 91.2%, outperforming traditional R-CNN and Fast R-CNN algorithms by more than 20%. Furuzuki et al. [15] utilized Faster R-CNN for liver tumor detection in abdominal CT images. Their approach first extracted the liver region as a preprocessing step, then applied Faster R-CNN to detect tumors within this region. Using ResNet50 and ResNet101 as backbone networks, they achieved average precision scores of up to 73.49%. Chen et al. [16] developed an improved Faster R-CNN model for detecting esophageal cancer in CT images. They introduced an online hard example mining (OHEM) mechanism to enhance detection accuracy. The model was trained on 1,520 gastrointestinal CT images from 421 patients. Their improved Faster R-CNN achieved an F-1 measure of 95.71% and a mean average

precision (mAP) of 92.15%, with a detection time of only 5.3 seconds per CT image.

These studies highlight Faster R-CNN's versatility and effectiveness in analyzing CT images across different anatomical regions and for various types of lesions.

D. Magnetic Resonance Imaging (MRI)

The application of Faster R-CNN in MRI analysis capitalizes on the modality's superior soft tissue contrast and multiplanar imaging capabilities, primarily encompassing the detection of cerebral tumors, cardiac function assessment, and musculoskeletal disease diagnosis. Avşar and Salçin [17] applied Faster R-CNN to detect and classify brain tumors in MRI images. Their model was trained on 3,064 MRI slices containing three types of tumors: meningioma, glioma, and pituitary. Using ResNet-50 as the backbone network, they achieved a classification accuracy of 91.66% on the test set. The model could simultaneously localize tumors and classify them into the three categories. Ma et al. [18] utilized Faster R-CNN for detecting cervical spinal cord injuries and disc degeneration in MRI scans. They trained models using both ResNet-50 and VGG-16 as backbone networks on a dataset of 1,500 cervical MRI images. The model achieved a mAP of 88.6%. Kermani et al. [19] developed a Faster R-CNN-based approach for heart localization and right ventricle wall motion abnormality detection in cardiac MRI. They proposed a novel architecture called NF-RCNN, which uses Network-in-Network as the region proposal network. Tested on two cardiac MRI datasets, their model achieved high accuracy in localizing the heart and detecting wall motion abnormalities.

The algorithm effectively manages the high-resolution and multi-modal characteristics inherent to MRI, facilitating enhanced comprehension of intricate anatomical structures and pathological alterations.

E. Ultrasound

In the field of ultrasonography, Faster R-CNN has been deployed to analyze real-time, operator-dependent images characterized by speckle noise and acoustic shadows, with applications. Bose et al. [20] applied an optimized Faster R-CNN for detecting lesions in breast ultrasound images. They conducted experiments on a dataset of 1,183 images. Wu et al. [21] developed a Faster R-CNN-based method for automatically recognizing five standard planes in fetal heart ultrasound images. They employed MobileNet as the backbone network and trained and tested the model on 1,500 images. The model achieved a mAP of 88.6% on the validation set, effectively identifying and localizing different cardiac planes. Sapitri et al. [22] utilized Faster R-CNN to detect aortic regions in fetal ultrasound images. They trained and tested the model on 151 images, employing data augmentation techniques to expand the training set. The method achieved a mAP value of 83.71% on the test set.

The algorithm's capacity for real-time processing of ultrasound images enhances both the accuracy and efficiency of examinations, particularly in clinical scenarios necessitating rapid decision-making.

F. Microscopy

Faster R-CNN is also utilized in microscopic image analysis, particularly in pathology and cytology, where it processes high-magnification, high-resolution images to identify cellular and subcellular structures. Zeng et al. [23] developed a Multi-data Faster R-CNN approach for detecting

white blood cells (WBCs) in hyperspectral microscopic images. They combined spectral data analysis using 1D-CNN with spatial image analysis using Faster R-CNN. The model achieved mAP of 0.962 and F1 scores above 0.9 for classifying five types of WBCs. Avci et al. [24] applied an optimized Faster R-CNN model to detect and classify urine sediments in microscopic images. They used super-resolution techniques to enhance low-quality images and employed denoising methods to improve image quality. The model was able to classify 10 different types of urine sediments with an accuracy of 98.6%. Rosati et al. [25] utilized Faster R-CNN for detection and quantification of DNA damage in comet assay images. Their model achieved a mean average precision of 0.74 in detecting different levels of DNA damage.

These studies highlight Faster R-CNN's versatility in handling various types of microscopic images, from blood cells to urine sediments and even subcellular structures.

G. Endoscopy

Faster R-CNN is applied in endoscopic image analysis to detect various abnormalities and artifacts in luminal structures, processing images characterized by variable illumination, specular reflections, and motion blur. Chen et al. [26] developed a self-attention based Faster R-CNN for detecting polyps in colonoscopy images. They incorporated an attention module to enhance feature representation and achieved 96.2% sensitivity and 76.2% specificity. Xia et al. [27] applied Faster R-CNN to detect gastric lesions in magnetically controlled capsule endoscopy images. Their model detected lesions with 96.2% sensitivity and 76.2% specificity across 5 lesion types. Yin et al. [28] utilized Faster R-CNN to detect 8 types of artifacts in endoscopy videos, including specularities, bubbles, and saturation. Their Faster R-CNN implementation achieved a detection score of 0.1827 on the EAD2020 dataset.

These studies demonstrate Faster R-CNN's effectiveness in analyzing various types of endoscopic images, from colonoscopy to capsule endoscopy, for both abnormality and artifact detection.

H. Optical RGB images

Faster R-CNN is also employed for medical diagnosis tasks using standard optical RGB images, which offer color information and wide field-of-view capturing, especially in the fields of ophthalmology and dermatology, with applications including diabetic retinopathy detection, skin disease classification, and burn wound analysis. Nur-A-Alam et al. [29] developed a Faster R-CNN-based method to detect diabetic retinopathy from retinal fundus images. They incorporated feature fusion and achieved 98.58% accuracy in classifying normal vs diabetic retinopathy images. Josphineleela et al. [30] applied Faster R-CNN for multi-class skin disease classification using dermoscopic images. Their model achieved 95.82% accuracy in classifying 8 types of skin lesions. Pabitha and Vanathi [31] utilized Faster R-CNN to analyze burn wounds from regular camera images. The model could classify burn degrees and suggest appropriate treatment methods with 98.46% precision.

These studies demonstrate Faster R-CNN's effectiveness in analyzing various types of medical optical images beyond specialized medical imaging modalities, from retinal and skin photographs to general wound images, for diagnostic purposes.

I. Other healthcare applications

Beyond traditional medical imaging applications, Faster R-CNN has demonstrated potential in several innovative domains. These include posture detection and analysis, medication recognition, food recognition for dietary monitoring and assistive technologies for the visually impaired. Lv et al. [32] used a Faster R-CNN approach for detecting traditional Chinese medicines in images. They used Faster R-CNN to identify and classify 100 common types of traditional Chinese medicines. The model achieved mAP of 85.56% and precision of 85.12% for classifying these medicines. Devi et al. [33] applied an optimized Faster R-CNN model combined with Inception-ResNet-V2 to detect objects and assist visually challenged people. They used multiple datasets including Indoor Scene and Human4 Dataset. The model was able to detect objects with 85.56% mAP and provide audio feedback. Cherpanath et al. [34] utilized Faster R-CNN for food image recognition and calorie prediction. Their model achieved 94.6% accuracy in detecting food items and estimating their calorie content. Zhang et al. [35] combined Faster R-CNN with MediaPipe for range of motion assessment in spinal diseases and frozen shoulder. Their system achieved 88.4% mAP.

These studies highlight Faster R-CNN's versatility in handling various types of medical imaging tasks, from drug identification to motion analysis and food recognition.

IV. CONCLUSION AND FUTURE DIRECTIONS

This review examines Faster R-CNN's applications in healthcare from 2017 to 2024, highlighting its effectiveness across various medical imaging modalities. However, challenges persist. A key limitation is medical datasets often have limited samples for specific conditions, and different datasets may exhibit significant feature variations, leading to insufficient transfer learning capabilities. Even within the same disease category, presentations can vary significantly across patients and imaging equipment. Most current research focuses on single-function models targeting specific conditions, which may limit broader applicability in clinical settings. Additionally, as models become more complex, maintaining interpretability of their decision-making processes becomes increasingly challenging, yet crucial for clinical trust and adoption.

Future research should address data limitations through advanced data augmentation techniques, including careful application of Generative AI for synthetic image creation. Improving generalization across datasets and equipment is important for clinical adoption. A shift towards comprehensive systems is emerging, focusing on integrated models for multiple conditions and personalized healthcare. These systems should incorporate multi-modal imaging data, electronic health records for holistic patient care. Other key areas include real-time processing for live procedures, federated learning for privacy-preserving training, and explainable AI techniques. As technology and medical knowledge progress, Faster R-CNN has the potential to play an increasingly significant role in precision medicine and personalized healthcare. By addressing current limitations and expanding capabilities, these advanced systems could potentially improve diagnostic accuracy, treatment planning, and patient outcomes across a wide range of medical specialties, moving towards a more personalized and efficient healthcare paradigm.

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